Technology in Society 82 (2025) 102874



Contents lists available at ScienceDirect

Technology in Society

journal homepage: www.elsevier.com/locate/techsoc





Digital transformation and total factor productivity in manufacturing firms: Evidence of corporate public responsibilities in China

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ARTICLE INFO

Keywords:
Digital strategy
Employee public responsibilities
Corporate productivity
Sustainable social development
China

ABSTRACT

Fulfilling corporate public responsibilities to employees is crucial for sustainable social development. The impact of digital transformation on total factor productivity (TFP) about these responsibilities remains unclear. Taking Chinese manufacturing companies as the research sample, empirical analyses including fixed effect model, propensity score matching, and staggered Differences-in-Differences methods have demonstrated that digital strategy can boost manufacturing companies' TFP. Further, stepwise regression analysis of the mechanisms revealed that while the transformation primarily boosts productivity directly, it also indirectly boosts TFP by reinforcing employee and public responsibilities. These findings suggest that maintaining responsibilities to employees and the public during digital transformation can enhance corporate productivity. This study sheds light on how digital technology drives corporate social responsibility for sustainable social development.

1. Introduction

Digital technologies are accelerating their integration and penetration into industries, driving changes in industrial organizational logic and systems, and promoting the transformation of business models towards information production and service-oriented models (Bianchini & Moritz, 2022; Ciarli et al., 2021; Kraus et al., 2022; Priyono et al., 2020). The main goal of digital transformation is to address challenges faced by companies in terms of efficiency and effectiveness. Some research has shown that implementing digital technologies can result in a favorable effect on a company's performance (Chen et al., 2016; Lu et al., 2024; Luo et al., 2018; Sousa-Zomer et al., 2020; Zhai et al., 2022), operations (Gupta et al., 2019), output (Pan et al., 2022), productivity (Cheng et al., 2023; Ding, Sheng, et al., 2024; Jin et al., 2023; Relich, 2017; Wang et al., 2023a), and innovation (Zhuo and Chen, 2023). However, in some cases, computerization, automation, or digitization technologies still struggle to improve productivity (Brown, 2015; Harris, 2001; Ribeiro et al., 2023).

"Fulfilling employee and public responsibilities" refers to the commitment and actions taken by a company to ensure the well-being and satisfaction of its employees, as well as to contribute positively to the communities and society in which it operates. This includes providing a safe and fair working environment, offering opportunities for professional development and growth, engaging in philanthropic activities, and minimizing negative impacts on the society. Previous research has predominantly investigated the effects of digital transformation on corporate social responsibility, emphasizing environmental and economic outcomes (Alkaraan et al., 2022; Chen & Hao, 2022; Ding, Sheng, et al., 2024; Ghobakhloo et al., 2023; Lin & Zhang, 2023; Zhao et al., 2024). However, these studies have largely neglected the dimensions of employee and public responsibility within the digital transformation context. By considering employee and public responsibilities, companies can create a more holistic approach to corporate social responsibility that benefits both internal and external stakeholders, leading to sustainable social development.

The primary objective of digital transformation implementation is to enhance organizational efficiency, with a lesser emphasis on fulfilling social responsibility (Ding, Sheng, et al., 2024). Digital technology

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https://doi.org/10.1016/j.techsoc.2025.102874
Received 9 July 2024; Received in revised form 25 January 2025; Accepted 10 March 2025
Available online 11 March 2025
0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

serves as a double-edged sword, as it can both facilitate the fulfillment of corporate social responsibilities and lead to irresponsible digitalization practices (Alexopoulos & Cohen, 2016; Bisht et al., 2023; Cardinali & De Giovanni, 2022; DeCanio, 2016; Kong & Liu, 2023). Companies can leverage digital technology for better fulfillment of employee and public responsibilities, but digital transformation brings new challenges in fulfilling these responsibilities (DeCanio, 2016; Horváth & Szabó, 2019). For instance, digital transformation requires workforce and financial resources, which may divert resources originally allocated to fulfill corporate social responsibilities. Resource constraints make it challenging to better fulfill employee and public responsibilities (Aupperle et al., 1985). The application of digital technology has the potential to enhance job engagement (Nadeem et al., 2024; Selimović et al., 2021), but it may also reduce low-skilled jobs, decreased employment opportunities, and increased unemployment risks (Dengler & Matthes, 2018).

The literature presents an ambiguous view on the role of employee responsibility and public responsibility in digital transformation. On the one hand, companies' investment in social responsibility contributes to improving their economic performance (Tang and Yang, 2022). On the other hand, allocating limited production resources to fulfill employee responsibility and public responsibility may reduce the efficiency of the company. For example, scandals and controversies related to employee and public responsibility can damage a company's reputation and impact its effectiveness (Nirino et al., 2021; Walsh et al., 2009). Nevertheless, our understanding of how digital transformation, which is linked with employee and public responsibility, impacts corporate TFP remains limited.

The mechanism of digital transformation affecting corporate employee and public responsibilities has not been fully revealed, and it is a gap in the literature. This study aims to fill this gap by examining the dual impact of digital transformation on both corporate social responsibility and TFP, offering an innovative perspective on responsible digitalization. This paper proposes two research questions related to the Chinese manufacturing companies.

1) Does the digital transformation impact corporate TFP ?
2) What role do employee responsibility and public responsibility play in the impact of digital transformation on TFP ?

To answer these research questions, this study selects manufacturing companies as research samples and develops hypotheses, which are then empirically tested. The reason we choose manufacturing is that manufacturing is the leading carrier of digital transformation in China, with more than half of manufacturing companies still in the process of partial transformation. This study investigates whether the digital transformation of Chinese manufacturing companies improves TFP and explores the mediating role of employee responsibility and public responsibility, which is of great practical significance for better implementing responsible digital strategies in companies.

This study explores an under-researched area, namely responsible digitalization. The research contributes to the existing literature in the following aspects. Firstly, our research is pioneering in identifying that incorporating employee and public responsibilities into digital strategies enhances corporate productivity. Although related researchers have explored the relationship between the digital economy and single-factor productivity (Hao et al., 2023; Lyu et al., 2023; Pan et al., 2022), there is a lack of explanation regarding the effectiveness of digital transformation in improving TFP. This article reveals that the employee responsibility and public responsibility are pathways through which digital transformation enhances a company's TFP. This finding responds to the academic controversy regarding the role of fulfilling employee and public responsibility in boosting productivity. Secondly, this paper proposes a new theoretical framework for responsible digitalization, offering new insights into the mechanisms by which digital transformation impacts corporate productivity. Thirdly, the uniqueness of this study resides at the nexus of digital transformation and corporate employee and public responsibility. This novel perspective shapes future research endeavors in the realm of responsible digitalization. Additionally, this study also provides actionable guidance for policymakers and industry leaders aiming to harmonize technological advancements with social responsibilities.

The article is divided into several sections: theoretical foundation and hypotheses, research design, results and analysis, discussion, and conclusion.

2. Theoretical foundation and hypotheses

2.1. CSR challenges brought about by digitization

Individuals rely on their work to meet diverse basic needs, where work serves not only as a primary source of income but also carries psychological values such as self-actualization, a sense of achievement, and happiness, as well as social benefits including social status, respect, social interaction, a sense of belonging, and friendship (Du et al., 2015). Given the comprehensive support work provides for individual needs, the potential disruptive impact of digital technology on the job market will extend far beyond the economic sphere, bringing about profound social and political consequences. Corporations have a responsibility to safeguard their employees' rights and interests, and thus should actively intervene to address the unemployment challenges caused by digital technology.

The application of digital technology has the potential to enhance job engagement (Nadeem et al., 2024; Selimović et al., 2021). Nadeem (Nadeem et al., 2024) emphasized that employees constitute invaluable assets during digital transformation, noting its substantial impact on the attitudes, psychology, and job-related aspects of personnel in Industry 4.0. Selimović et al. (Selimović et al., 2021) discovered that interpersonal connectedness in digital work settings notably influences employee performance and well-being, boosting their inclination to embrace digital workplace transformation. Digitalization alters the skill demands on company employees (Basl, 2017; Gekara & Thanh Nguyen, 2018; Müller & Voigt, 2018). The integration of smart computers and robotics in the workplace has shifted job requirements and organizational structures, emphasizing the need for soft, generic, and transferable skills, particularly computer proficiency, in a highly mechanized and digitalized work environment.

Digital technology and its application may result in a decline in low-skilled job roles, reduced employment opportunities, heightened unemployment risks, and increased workforce mobility (Dengler & Matthes, 2018). Gekara and Thanh Nguyen (Gekara & Thanh Nguyen, 2018) considered that employees lacking relevant training may find it challenging to execute their job responsibilities, thereby elevating unemployment risks. Furthermore, digital transformation introduces novel challenges and responsibilities for employees (Horváth & Szabó, 2019).

As a pivotal digital technology, AI offers personalized services to humans, significantly enhancing work efficiency across diverse sectors (Dwivedi et al., 2021). Researchers have established that AI applications can automate repetitive tasks such as data entry, invoice processing, and online customer service (Ribeiro et al., 2021). These expert systems excel in rapid data analysis, optimizing workflows, and making complex decisions with greater speed and accuracy than humans, thereby boosting efficiency and productivity in various industries (Javaid et al., 2021; Ng et al., 2021; Wamba-Taguimdje et al., 2020). AI-driven chatbots and virtual assistants provide round-the-clock, customer-centric personalized recommendations (Camilleri & Troise, 2023; Selamat and Windasari, 2021), enabling employees to focus on higher-value tasks.

Digital technology has the potential to exacerbate social inequalities, as job role reassignments and duty redefinitions may challenge employees' ability to adapt to new processes (Budhwar et al., 2023). André et al. (André et al., 2018) revealed that AI application can impair individuals' capacity to make autonomous choices (André et al., 2018). Furthermore, AI-enabled products have detrimental effects on

autonomy and wellbeing, manifesting in various forms of digital addiction (Samaha and Hawi, 2016). Increasing evidence suggests that digital addiction is widespread and adversely affects academic performance, relationship quality, and life satisfaction (Roberts and David, 2016; Samaha and Hawi, 2016). Failure to address these challenges adequately during the transformation process can impair corporate efficiency and productivity.

Scholars have examined the impact of digitization on corporate social responsibility (CSR) across diverse geographical contexts and industries, extending beyond China. For instance, based on survey samples obtained from countries such as Denmark, Finland, Sweden, the United States, Singapore, Taiwan, Switzerland, Norway, the Netherlands, and the United Arab Emirates, studies have found that the application of digital technology has the potential to boost job engagement (Nadeem et al., 2024; Selimović et al., 2021). Basl (Basl, 2017) and Dengler (Dengler & Matthes, 2018) observed that digitalization modifies the skill requirements for employees in Czechia and Germany, and it also increases unemployment risks in Germany. When comparing the conclusions of relevant studies across different industries, research based on financial institutions, banks, and insurance companies has revealed that interpersonal connectedness in digital work environments significantly affects employee performance and well-being (Selimović et al., 2021). Additionally, studies have investigated the influence of artificial intelligence on the performance of large technology firms (Wamba-Taguimdje et al., 2020). In the Chinese context, scholars have discovered that digital transformation enhances CSR performance by improving green management innovation and analyst coverage (Sun et al., 2023). Furthermore, digitalization positively impacts CSR in the Chinese mining industry (Xu et al., 2023).

In conclusion, digital technology is a double-edged sword, and if used effectively, it can benefit humanity. However, if the challenges posed by digital technology are not properly controlled and appropriate measures are not taken, the possession of digital technology may result in more harm than good. Despite some scholars having explored the relationship between digital transformation and corporate social responsibility, empirical research based on the Chinese manufacturing industry remains insufficient. These studies have failed to elucidate how digital transformation impacts employee and public responsibilities within the framework of corporate social responsibility.

2.2. Corporate social responsibility and TFP

Stakeholder theory emphasizes the interaction and cooperation between organizations and external stakeholders. By balancing the needs and interests of various stakeholders, organizations can establish more stable and favorable relationships, achieving a win-win situation. When fulfilling social responsibilities, enterprises should consider the interests and needs of all stakeholders related to the enterprise. Through dialogue and cooperation with stakeholders, enterprises can comprehensively consider the internal and external economic, ecological, and social environments of the enterprise, and formulate more comprehensive and effective social responsibility strategies.

The discussion on the relationship between CS and TFP has been ongoing. From the literature, the explanation for how CSR enhances TFP is multidimensional. TFP reflects, in part, the contribution of technological progress to economic growth. From the perspective of corporate innovation, technological progress is a direct source of TFP growth, and technological progress cannot be achieved without corporate innovation activities. Some studies have found that corporate environmental and social performance helps to amplify the quality and efficiency of corporate innovation (Chen et al., 2023; Zhou et al., 2023), and also promotes green collaborative innovation outputs among enterprises (Wang et al., 2023b; Wu et al., 2023), ultimately improving corporate TFP (Ding, Sheng, et al., 2024). Adopting socially responsible practices enables firms to forge robust relationships with key stakeholders (Ullah and Sun, 2021), thereby enhancing their innovation capabilities for new technologies, products, and markets (Branco & Rodrigues, 2006). Furthermore, these practices foster workforce loyalty and reduce absenteeism, ultimately elevating firm productivity (Faleye & Trahan, 2011).

Firms can enhance TFP not only through technological advancements but also by optimizing the allocation of production factors with digital technologies. Misallocation of resources can impact a firm's R&D and market decisions (Peters and Romi, 2014), subsequently hindering productivity improvements (Liang et al., 2022). Due to financing constraints, when firms lack sufficient funds, they often forgo advantageous investment opportunities, affecting optimal production decisions and ultimately leading to distorted resource allocation. Companies with good CSR performance may obtain various resources and face lower capital constraints (Cheng et al., 2014). Scholars have discovered that CSR initiatives decrease capital costs by alleviating agency costs and asymmetric information (El Ghoul et al., 2011; Oikonomou et al., 2014), suggesting that heightened CSR leads to improved resource allocation. Firms deeply involved in CSR experience notably lesser capital constraints due to lower funding costs from debtholders (Cheng et al., 2014). By minimizing market frictions and agency costs, socially responsible firms enjoy a superior position in securing financing in capital markets (Samet and Jarboui, 2017). On the other hand, enterprises establish good government-enterprise relationships through CSR, which facilitates their access to government grants (Khurshid et al., 2024). He et al. (He et al., 2023) proposed that subsidies can be viewed as a tool to address environmental externalities and promote CSR implementation. However, Khurshid emphasized that subsidy policies primarily serve to generate funds for private enterprises, enhancing their financial capacity. Therefore, it can be known from the literature that CSR can improve the total factor productivity.

2.3. Digital transformation and corporate productivity

Endogenous growth theory posits that technological progress is an endogenous variable and asserts that economic growth is sustained through continuous technological advancements (Romer, 1990). TFP measures how effectively input factors are converted into final output (Syverson, 2011). The integration of digital technologies with industries has driven the transformation of organizational logic and systems, promoting the shift from existing development models to those centered around information production and information services (Franke & Von Hippel, 2003; Hinings et al., 2018; Liu et al., 2011; Rammer et al., 2022; Teece, 2018). The theory suggests that increased productivity can be achieved through the development of technologies, the acquisition of knowledge, and the nurturing of talents.

The allocation of production resources is a key factor affecting productivity (Jiang & Li, 2024). Studies have found that the misallocation of production factors leads to low productivity (Adamopoulos et al., 2022; Brandt et al., 2013). When data as a factor of production is integrated with digital tools, companies can strengthen the allocation of production resources and maximize the efficiency of input (Ferreira et al., 2019; Heavin & Power, 2018). In addition, data not only carries information but also knowledge, experience, and skills. Digital technology improves the capacity for facilitating knowledge sharing and empowering skills (Jiang & Li, 2024), which can boost organizational transparency and collaborative innovation, ultimately leading to better utilization of corporate resources (Ding, Vuković, et al., 2024). Based on the above analysis, this paper proposes that digital transformation can enhance productivity through resource reallocation.

Corporate performance is closely linked to corporate TFP. Integrating digital technology into product design, processes, and organizational structure can boost operational efficiency and enhance the organization's performance (Chen et al., 2016; Gupta et al., 2019; Luo et al., 2018; Porter and Heppelmann, 2014; Rammer et al., 2022; Sousa-Zomer et al., 2020; Tortorella et al., 2020; Yunis et al., 2018). Furthermore, digitalization plays a crucial role in enhancing regional

productivity (Aly, 2020; Gaglio et al., 2022; Pan et al., 2022). Given that firms serve as vital vehicles for boosting regional TFP, we propose the following hypotheses 1.

H1. Digital transformation can enhance a company's TFP.

2.4. Digital transformation, employee responsibility, and TFP

The stakeholder theory holds that businesses are responsible to shareholders, such as employees, the environment, suppliers, customers, and consumers, to maximize overall benefits. As a key component of the CSR, employee responsibility refers to a firm's fulfillment of its social responsibility towards its employees, reflecting the protection of their legitimate rights and interests. Specifically, corporate employee responsibilities include employee training, safety training, employee welfare support, and improving employee benefits.

Investing in employee responsibilities can improve corporate TFP. For example, training employees can enhance their skills while showing concern for employees and improving their wages and benefits through social responsibility initiatives, which can foster a sense of belonging and loyalty and stimulate their initiative, proactivity, and creativity. Additionally, human resources are essential productive factors (Yang and Xiao, 2024). The better the development and utilization of human resources, the more likely it is to promote an increase in TFP. Demonstrating good performance in employee responsibilities can also attract talent, helping retain high-quality, highly skilled employees and attract talented professionals. The above analyses indicate that the better the fulfillment of employee responsibilities, the higher the TFP.

The consequences of technology depend on the background and period in which it interacts with the market, society, and the environment (Dedrick et al., 2013; Kallal et al., 2021; Kewell et al., 2017; Su et al., 2022). Adopting digital technology can also lead to irresponsible digitalization (Cardinali & De Giovanni, 2022; De Giovanni, 2021). Although digital technology has the potential and advantages of reducing operational risks and improving efficiency, it also reduces some employment opportunities (Dengler & Matthes, 2018; Zhong et al., 2024). Digitalization changes company employees' skill requirements (Basl, 2017; Gekara & Thanh Nguyen, 2018; Müller & Voigt, 2018). Employees who do not receive relevant training may struggle to perform their job duties, increasing the risk of unemployment (Gekara & Thanh Nguyen, 2018). Therefore, digital technology and its application can lead to a reduction in low-skilled jobs, a decrease in employment opportunities, and an increase in the risks of unemployment and workforce mobility. Digital transformation also brings new challenges and tasks to fulfill employee responsibility (Horváth & Szabó, 2019). If these challenges and issues are not properly handled during the transformation process, they will damage corporate efficiency.

Digital transformation primarily impacts corporate social responsibility by effectively integrating various resources, including the enhanced coordination of human capital and employment opportunities (Jiang & Li, 2024; Sun et al., 2024). Cardinali et al. (Cardinali & De Giovanni, 2022) found that a combination of digital technologies helps companies achieve responsible digitalization. We infer that the responsible digitalization is also reflected in the responsibility of employees. Although digital transformation poses challenges to fulfilling employee responsibilities in companies, employee responsibility may be a key pathway for digital transformation to improve TFP. Therefore, we proposes hypothesis 2.

H2. Digital transformation improves TFP in companies by promoting the fulfillment of employee responsibilities.

2.5. Digital transformation, public responsibility, and TFP

Companies' survival and development rely on various stakeholders' participation, and companies need to obtain key resources from stakeholders through resource exchange. As public expectations for public responsibility increase, charity and donations have gradually become important means for companies to improve and maintain stakeholder relationships, acquire key resources, and enhance corporate performance (Wang and Qian, 2011). Engaging in charitable activities is an important manifestation of corporate public responsibility, as it helps establish a positive image and reputation for the company, making various stakeholders more willing to cooperate and provide production resources. Fulfilling public responsibilities brings about a more abundant production factor, which helps optimize the allocation of production factors and improve TFP.

Companies face pressures from various stakeholders, such as governments and society, which require them to pay attention to public responsibility issues. Digitalization enhances transparency and promotes corporate accountability in fulfilling public responsibilities. Information sharing and integration during digital transformation reshape resource allocation and facilitate public responsibilities. For example, the combination of digitalization and public welfare enables faster dissemination of public welfare information, broader coverage of public welfare, and lower costs for conducting public welfare activities, making it easier for companies to fulfill their public responsibilities, such as donations. Based on the analysis, we can conclude that digital transformation enhances the efficiency of fulfilling corporate public responsibilities. By fulfilling these responsibilities, companies can more easily obtain abundant factors of production. This helps to further promote the improvement of TFP. Therefore, hypothesis 3 is proposed.

H3. Digital transformation can help companies fulfill their public responsibilities, thereby enhancing TFP.

The mechanism digital transformation enhances the productivity of manufacturing companies is illustrated in Fig. 1.

3. Research design

Upon completing data collection for this study, we quantified the key variables. We quantified digital transformation intensity based on the web crawler technology. To estimate TFP and ensure the reliability of our findings, we utilized several methods, including LP, FE, OLS, and GMM. Subsequently, a fixed-effects model is used to examine the main effect. Furthermore, endogeneity tests and related tests are conducted to ensure robustness. Additionally, we employed the stepwise regression method to shed light on the role of employee and public responsibility. The use of these methods can provide reliable conclusions as far as possible.

3.1. Samples and data

This article selects manufacturing companies listed on the Shanghai and Shenzhen stock exchanges from 2010 to 2022. The financial and corporate governance data involved in the research were obtained from corporate annual reports and the CSMAR database. Employee and public responsibility data are sourced from the CSR ratings of listed companies published on Hexun.com. The reason for selecting 2010 as the study's starting point is that Hexun.com began disclosing CSR scores in 2010. ST

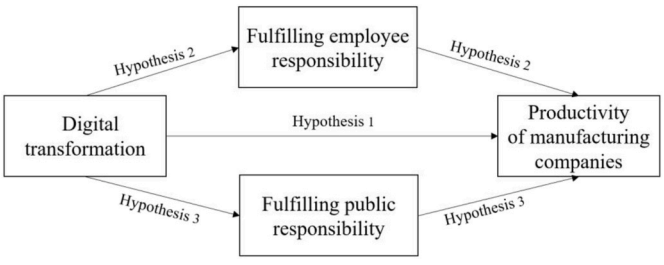


Fig. 1. Interactions between digital transformation, fulfilling responsibility, and TFP.

and PT companies were excluded from the sample, along with data containing abnormalities or omissions. Finally, 17,032 valid observations were obtained.

3.2. Methodology

The main statistical methods used in this study include the fixed effects model and the mechanism analysis method proposed by Alberto Alesina (Alesina & Zhuravskaya, 2011). To increase the robustness of the research results, this study also employs the Difference-in-Differences (DID) method, Propensity Score Matching (PSM) method, and Generalized Method of Moments (GMM) for testing.

3.2.1. Fixed effects model

Fixed effects model is a commonly used panel data analysis method in statistics. In panel data, we have observations of the same group of individuals at multiple time points. The basic idea is to control individual heterogeneity that does not change over time but may affect the dependent variable by introducing one or more fixed effects representing individual characteristics, in order to estimate the impact of explanatory variables more accurately on the dependent variable.

3.2.2. Mechanism analysis method

This article presents several regression models to test research hypotheses. Model (1) is constructed to validate H1. Models (2) and (4) are used to validate H2. Models (3) and (5) are used to validate H3. Mechanism variables are public responsibility (*Pucsr*) and employee responsibility (*Emcsr*). *Control* represents control variables. The regression analysis in the article not only adopts a fixed effects model but also includes year (*Year*) and industry (*Industry*) as control variables and re-estimates the regression equations. Stata 17 is used as the statistical analysis software for empirical analysis.

$$TFP_{i,t+1} = \alpha_0 + \alpha_1 Transformation_{i,t} + \sum_m \alpha_m Control_{it} + \varepsilon_{it} \quad (1)$$

$$Emcsr_{i,t} = \beta_0 + \beta_1 Transformation_{i,t} + \sum_j \beta_j Control_{it} + \varepsilon_{it} \quad (2)$$

$$Pucsr_{i,t} = \beta_0 + \beta_1 Transformation_{i,t} + \sum_j \beta_j Control_{it} + \varepsilon_{it} \quad (3)$$

$$TFP_{i,t+1} = \gamma_0 + \gamma_1 Transformation_{i,t} + \gamma_2 Emcsr_{it} + \sum_k \gamma_k Control_{it} + \varepsilon_{it} \quad (4)$$

$$TFP_{i,t+1} = \gamma_0 + \gamma_1 Transformation_{i,t} + \gamma_2 Pucsr_{it} + \sum_k \gamma_k Control_{it} + \varepsilon_{it} \quad (5)$$

This paper draws on the mechanism analysis method proposed by Alberto Alesina (Alesina & Zhuravskaya, 2011) to analyze employee and public responsibility roles. This mechanism analysis method is scientific and widely applicable, and it becomes the preferred tool for academic research. The primary idea of mechanism analysis is to determine whether the addition of the mechanism variable leads to changes in the coefficient of the mechanism variable, thus verifying the existence of the mechanism. The regression of variable *CSR* (*Emcsr* or *Pucsr*) on *Transformation* is first conducted to verify that *Transformation* statistically affects the mechanism variable *CSR*, providing initial evidence for the subsequent paths of influence of *Transformation*, *CSR*, and *TFP*. Next, the research examines the coefficient variations in two scenarios: one where the CSR variable is controlled for (that is, regressing *TFP* on *CSR* and *Transformation*) and another where it is not (that is, regressing *TFP* on *Transformation*). This analysis aims to confirm the presence of the mechanism.

3.2.3. Difference-in-differences method

The basic idea of DID method is to estimate the effect of an event by comparing the difference in changes between the treatment group and the control group before and after the event implementation. Specifically, it calculates the difference between the average change in the treatment group after the event and the average change in the control group during the same period. The steps of DID method are as follows: first, identify the treatment group and control group, and collect relevant data on them before and after the digital transformation. Second, calculate the differences. Calculate the difference between the treatment group and control group before the transformation. Then, calculate the difference between the treatment group and control group after the transformation, subtracting the result of the first difference to obtain the double difference estimate. Third, estimate the treatment effect and its significance. It is worth noting that the effectiveness of the DID method relies on the parallel trends assumption, which states that before the transformation event, the outcome variables of the treatment group and control group should have the same time trend.

3.2.4. Propensity score matching (PSM) method

The PSM method is a widely used statistical approach for addressing causal inference issues. PSM helps to reduce interference between the dependent and independent variables by matching the propensity scores of the treatment and control groups. This leads to a more accurate estimation of treatment effects. The propensity score represents the likelihood of a company receiving a specific treatment given certain conditions. By calculating propensity scores for each company, PSM can match companies with similar scores in the treatment and control groups, thereby minimizing the impact of confounding factors on treatment effect estimation. The implementation steps of PSM involve calculating propensity scores using a Probit regression model, matching treatment and control groups based on these scores, assessing the quality of matching through balance tests, and estimating treatment effects.

3.2.5. Generalized Method of Moments (GMM)

GMM is a parameter estimation technique that relies on the premise that the true parameters of a model satisfy specific moment conditions. The core concept of GMM estimation involves leveraging the relationship between sample moments and population moments to estimate the model parameters. In the context of random sampling, assuming the model assumptions hold, it is always feasible to identify multiple moment conditions that the actual parameters adhere to. GMM estimation leverages these moment conditions to formulate the estimator. Initially, a moment function of the model parameters is defined, and the parameter estimates are derived by minimizing the weighted sum of squares of this moment function. In the regression analysis conducted in this study, GMM estimation was employed to enhance the robustness of the research findings.

3.3. Measurements of variables

3.3.1. Digital transformation

Blockchain, cloud computing, big data, and artificial intelligence are the underlying digital technologies. The first level of digital transformation focuses on the integration of digital technology. Furthermore, the transformation will gradually extend to companies' core and extended business. At this stage, companies pay more attention to integrating technology with business scenarios, transitioning from the first level to a more profound transformation of their business. This article separates digital transformation into two parts - "digital technology" and "application of digital technology."

Information related to digital transformation is reflected in corporate annual reports. The express in annual report reflects the development concepts and strategy of the companies, demonstrating their transformation practices, paths, and solutions. Referring to existing research, it is feasible and scientific to measure it based on frequency statistics of the term "digital transformation" in annals reports (Luo et al., 2023; Wen et al., 2022; Wu et al., 2022; Zhai et al., 2022). To extract the related term, we used the Java PDFbox library, which helped us filter

feature words and calculate frequencies. We refer to the feature words adopted by Ding (Ding, Sheng, et al., 2024), shown in Fig. 2. Utilizing the terms outlined in Fig. 2, we performed searches, comparisons, and tallied frequencies, while disregarding any phrases containing negations. The outcome is the frequency of related terms used as proxy indicators for digital transformation intensity.

3.3.2. Total factor productivity

TFP represents the overall efficiency of utilizing all production factors (including labor, capital, technology, etc.) in production. Corporate TFP is essentially a measure of resource allocation efficiency. Based on existing studies, Levinsohn-Petrin (LP) is a commonly used approach for estimating corporate TFP. This paper calculated the TFP based on the LP method, and multiple methods were used for the robustness test.

3.3.3. Employee responsibility and public responsibility

The CSR rating published by Hexun.com is currently the main way for the public to understand the employee responsibility and public responsibility of listed companies in China. This rating evaluates CSR in terms of employee responsibility, public responsibility, and others, and can reflect the CSR performance more comprehensively. In the Hexun rating, quantitative indicators of employee responsibility include average employee income, employee training, safety inspections and training, and care for employees. Public responsibility mainly refers to the contribution value to the public, with quantifiable indicators including the ratio of income tax to total profits and charitable donations.

3.3.4. Control variables

Referring to research by Liu (Liu et al., 2021), Zhai (Zhai et al., 2022), and Cui et al. (Cui et al., 2022), we have selected certain control variables for our study. These variables include the firm size (*Size*), leverage (*lev*), turnover of total capital (*ato*), operating income growth (*growth*), CEO Duality (*dual*), age (*firmage*), concentration (*balance*), share ratio of institutional investors (*inst*), management expense ratio (*mfee*), return on equity(*roe*), the company was audited (*audit*).

4. Empirical results and analysis

4.1. Regression analysis

Table 1 displays the regression outcomes that measure the relationship between digital transformation and TFP. In Regression (1), the results are obtained based on the fixed effects model while Regression (2) presents the outcomes after controlling "*year*" effect and "*industry*" effect. *Transformation* coefficients (0.0266; 0.0520) are both significantly positive. These regression outcomes support the H1.

Table 1
Baseline regression results.

Variables	(1)	(2)
	TFP	TFP
Transformation	0.0266***	0.0520***
	(7.75)	(17.16)
size	0.4471***	0.5953***
	(64.04)	(173.44)
lev	0.0603**	0.1038***
	(2.48)	(5.71)
ato	0.7786***	1.1153***
	(59.35)	(131.85)
growth	0.0218***	0.0155***
	(12.05)	(7.35)
dual	0.0127	0.0054
	(1.59)	(0.82)
firmage	0.0433	−0.0122
	(1.44)	(-1.31)
balance	0.0004	−0.0044
	(0.05)	(-0.89)
inst	0.0808***	0.0475***
	(4.97)	(3.33)
mfee	−0.6189***	−0.0023
	(-12.51)	(-1.00)
roe	0.0830***	0.1429***
	(5.92)	(9.00)
audit	0.0488*	0.0087
	(1.87)	(0.55)
Cons	−2.2492***	−5.6052***
	(-14.11)	(-73.89)
Industry	No	Fixed
Firm	Fixed	No
Year	Fixed	Fixed
Observations	17032	17032
R-squared	0.584	0.8244
F-statistic	924.371***	6644.840***

Note: t statistics in parentheses. *p < 0.1, **p < 0.05, ***p < 0.01.

4.2. Mechanism analysis

This study adopts the mechanism analysis method proposed by

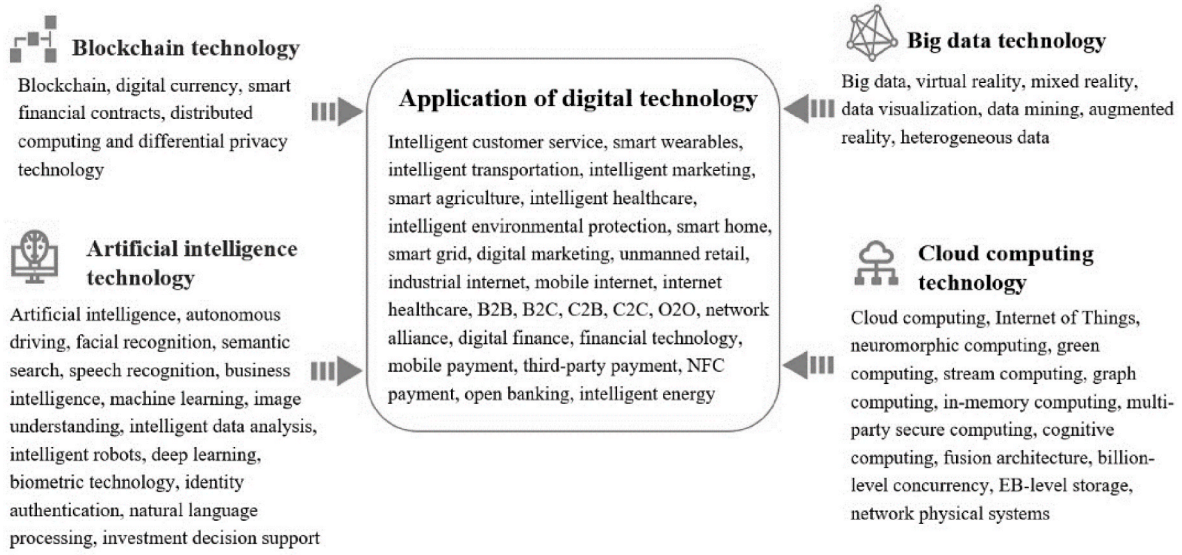


Fig. 2. Feature words of digital transformation.

Alberto Alesina (2011). The approach involves adding mechanism variables and observing the changes in the coefficient of the core explanatory variable to verify the existence of the mechanism.

Regression coefficients β for *Transformation* in regressions (1) and (3) are shown in Table 2. Both coefficients are positive at 5 % significance level with values of 0.0586 and 0.0763. In regressions (2) and (4), the coefficients for *Emcsr* ($\gamma_2 = 0.0036$; 0.0061) and Transformation ($\gamma_1 = 0.0263$; 0.0516) are significantly positive with a confidence level of 99 %. To ascertain the presence of a mechanism, the changes in the coefficients of *Transformation* were analyzed by adding the variable *Emcsr* to the baseline regression. As shown in Table 2, when *Emcsr* is added, the coefficient of *Transformation* changes from 0.0266 to 0.0263 in regression (2); the coefficient of *Transformation* changes from 0.0520 to 0.0516 in regression (4). The decreased coefficient suggests that *Emcsr* has a positive impact on the main effect. The findings from the regression analysis confirm H2.

Then, we introduce the variable *Pucsr* into the baseline regression presented in Table 1 and analyze the variations in the *Transformation* coefficients to assess the mechanism's presence. When *Pucsr* is added, the coefficient of *Transformation* changes from 0.0266 to 0.0265 in regression (2); the coefficient of *Transformation* changes from 0.0520 to 0.0516 in regression (4), as shown in Table 3. The regression findings offer backing for H2, indicating that digital transformation boosts TFP by elevating employee responsibility, as evidenced by the reduction in the *Pucsr* coefficient.

Through the analysis of the above results, it can be found that after adding mechanism variables, the *Transformation* variable remains significantly positive, but the regression coefficients have decreased considerably. This indicates that both the *Emcsr* and *Pucsr* variables are mechanism variables. The coefficient of the mechanism variable changes less, indicating that the transformation indirectly boosts TFP by strengthening employee and public responsibility practice, but mainly directly affects TFP.

Table 2
Regression results for testing the role of employee responsibility.

Variables	(1)	(2)	(3)	(4)
	Emcsr	TFP	Emcsr	TFP
Transformation	0.0586**	0.0263***	0.0763***	0.0516***
	(2.30)	(7.69)	(3.67)	(17.01)
Emcsr		0.0036***		0.0061***
		(3.21)		(5.44)
size	0.7011***	0.4446***	0.6873***	0.5911***
	(13.48)	(63.31)	(29.15)	(168.21)
lev	0.6771***	0.0579**	−0.5164***	0.1069***
	(3.74)	(2.38)	(-4.14)	(5.89)
ato	0.3999***	0.7772***	0.2632***	1.1137***
	(4.09)	(59.22)	(4.53)	(131.69)
growth	0.0171	0.0217***	0.0175	0.0154***
	(1.27)	(12.02)	(1.21)	(7.30)
dual	−0.1308**	0.0132*	−0.1566***	0.0063
	(-2.19)	(1.65)	(-3.48)	(0.96)
firmage	1.6263***	0.0375	0.4193***	−0.0147
	(7.25)	(1.24)	(6.57)	(-1.58)
balance	−0.1149*	0.0008	0.1015***	−0.0050
	(-1.94)	(0.10)	(2.98)	(-1.01)
inst	0.6578***	0.0784***	0.8247***	0.0425***
	(5.43)	(4.82)	(8.43)	(2.98)
mfee	2.1957***	−0.6268***	0.0125	−0.0024
	(5.96)	(-12.66)	(0.77)	(-1.03)
roe	0.0650	0.0827***	0.1500	0.1420***
	(0.62)	(5.91)	(1.38)	(8.95)
audit	0.0441	0.0486*	0.6678***	0.0047
	(0.23)	(1.87)	(6.15)	(0.30)
Constant	−16.6508***	−2.1896***	−14.3169***	−5.5183***
	(-14.03)	(-13.65)	(-27.47)	(-71.24)
Year	Fixed	Fixed	Fixed	Fixed
Industry	No	No	Fixed	Fixed
Firm	Fixed	Fixed	No	No
Observations	17032	17032	17032	17032
R-squared	0.136	0.584	0.186	0.849
F-statistic	103.372***	885.199***	168.812***	6146.300***

Table 3
Regression results for testing the role of public responsibility.

Variables	(1)	(2)	(3)	(4)
	Pucsr	TFP	Pucsr	TFP
Transformation	0.0617*	0.0265***	0.0926***	0.0516***
	(1.76)	(7.76)	(3.525)	(17.03)
Emcsr				
Pucsr		0.001		0.005***
		(0.69)		(5.12)
size	0.6448***	0.4467***	0.3896***	0.5935***
	(9.00)	(63.81)	(13.11)	(172.19)
lev	−1.7166***	0.0612**	−1.5603***	0.1109***
	(-6.88)	(2.52)	(-9.92)	(6.09)
ato	0.8566***	0.7781***	0.9712***	1.1109***
	(6.36)	(59.23)	(13.26)	(130.75)
growth	0.0641***	0.0217***	0.0735***	0.0152***
	(3.46)	(12.02)	(4.02)	(7.19)
dual	−0.0274	0.0127	−0.0072	0.0054
	(-0.33)	(1.59)	(-0.13)	(0.82)
firmage	0.0855	0.0433	0.4099***	−0.0140
	(0.28)	(1.44)	(5.09)	(-1.51)
balance	−0.1976**	0.0005	−0.1520***	−0.0037
	(-2.42)	(0.06)	(-3.54)	(-0.75)
inst	−0.0492	0.0808***	−0.1323	0.0481***
	(-0.30)	(4.97)	(-1.07)	(3.38)
mfee	−1.1133**	−0.6183***	−0.0178	−0.0023
	(-2.19)	(-12.50)	(-0.87)	(-0.96)
roe	2.0042***	0.0818***	2.2852***	0.1325***
	(13.95)	(5.81)	(16.62)	(8.29)
audit	−0.8171***	0.0492*	−0.2776**	0.0100
	(-3.06)	(1.89)	(-2.03)	(0.63)
Constant	−9.5940***	−2.2438***	−5.8865***	−5.5785***
	(-5.87)	(-14.061)	(-8.96)	(-73.42)
Year	Fixed	Fixed	Fixed	Fixed
Industry	No	No	Fixed	Fixed
Firm	Fixed	Fixed	No	No
Observations	17032	17032	17032	17032
R-squared	0.044	0.584	0.183	0.849
F-statistic	30.615***	884.170***	84.628***	6144.840***

4.3. Robustness checks

4.3.1. Analysis based on the DID method

Difference-in-differences (DID) method is a commonly used econometric method to evaluate the impact of policies, projects, or other interventions on a certain variable. This method compares the differences before and after intervention and between the treatment and control groups to determine the intervention's effect. The gradual and phased promotion of digital transformation by companies serves as an excellent quasi-natural experiment, which aligns with the scenario for applying the DID method. Therefore, this paper uses this staggered DID model to enhance the robustness of the research findings.

$$TFP_{i,t} = \alpha_0 + \alpha_1 \left(TreatG_i \times TreatP_{i,t}\right) + \sum \varphi_{it} Controls_{i,t} + \vartheta_t + \gamma_i + \varepsilon_{it} \quad (6)$$

Among them, $TreatG_i$ is a classified variable for the treatment group. $TreatP_{it}$ is a classified variable for the treatment period. γ_i represents fixed effects at the firm level, while ϑ_t represents fixed effects at the year level. This paper selects only firms that have completed the transformation process for a minimum of 5 years in a row. Besides, samples that consistently exhibit feature vocabulary are excluded.

Before conducting the DID analysis, we confirmed the validity of the parallel trend assumption. Model (7) represents the regression model for the parallel trends test. If a company i implemented a transformation strategy during the k-j period, the variable assignment is 1; otherwise, it is 0. *A* represents the period after the transformation, while *B* represents the period before. The year prior to the transformation is designated as

the control group.

$$TFP_{i,t} = \beta + \sum_{j=-B}^{A} \theta_j Transformation_{i,k-j} + \sum \varphi Controls + \sum Year + \varepsilon \quad (7)$$

This study adopts Stata 17 software to conduct parallel trend tests. As shown in Fig. 2, before digital transformation, the coefficient of θ fluctuates around 0 with a confidence interval that includes 0. However, this coefficient experienced a notable increase in the years of transformation and subsequent years. Before the transformation, the change in TFP between the treatment and control groups was not significantly different, indicating that the assumption of parallel trends is met. After implementing digital transformation, there has been a noticeable increase in TFP, as shown in Fig. 3.

Because the parallel trends assumption is satisfied, regression model (4) is used for regression analysis. Table 4 presents the results of the DID test. The regression coefficient of the *TreatG*×*TreatP* ($\alpha_1 = 0.051$, $p < 0.01$) is significantly positive, indicating a significant improvement in corporate TFP after digital transformation. From the results, it can be determined that the core conclusion remains unchanged after addressing endogeneity issues.

Table 4
Regression results by using the staggered DID method.

Variables	*TFP*	
	Coefficient	T-value
TreatG × *TreatP*	0.051***	5.04
size	0.471***	32.29
lev	0.083**	1.96
ato	0.727***	23.16
growth	0.095***	7.02
dual	0.011	0.98
firmage	0.073*	1.73
balance	−0.020	−1.37
inst	0.078***	3.44
mfee	−0.717***	−3.90
audit	0.051	1.46
Constant	−2.087***	−6.25
Year	Fixed	–
Firm	Fixed	–
Observations	13616	–
R-squared	0.932	–
Within R-sq.	0.4581	–
F-statistic	208.517***	–

4.3.2. Estimation by using GMM method

Drawing on the research of Du et al. (Du et al., 2023), we conducted a robustness test using GMM method. In the regression, the lagged one-period variables of the dependent variable were added to the dynamic panel model. Table 5 presents the results based on the GMM regression.

In these five Models, the Arellano-Bond test for AR(1) is significant at a 1 % level, suggesting the need to include $TFP_{i,t-1}$, $Emcsr_{i,t-1}$, or $Pucsr_{i,t-1}$ in the models. However, AR(2) is not significant in these five Models. The Hansen J test yields P value of 0.133, 0.137, 0.159, 0.136, 0.123, respectively, under heteroscedasticity assumption. This indicates that we cannot reject the null hypothesis that all attack variables are exogenous. Each coefficient of Transformation is significant at the 5 % level, as determined by GMM estimation.

The test results based on GMM indicate that when the *Emcsr* variable (or *Pucsr*) is added, the coefficient of *Transformation* decreases. This reduction in the coefficient suggests that both *Emcsr* and *Pucsr* have a positive impact on the main effect. The findings from the GMM regression analysis still confirm our hypotheses.

4.3.3. Regression by replacing methods for estimating the explanatory variables

We used OLS, GMM, and FE methods to calculate TFP and tested again. Table 6 show that all *Transformation* coefficients are significantly positive with a confidence level of 99 %, which confirms that our hypothesis 1 still holds.

4.3.4. Estimation based on Driscoll-Kraay standard error

Panel data combines the properties of cross-sectional and time series, so heteroscedasticity, serial, and cross-sectional correlation may exist in panel data. This paper uses Driscoll-Kraay's (Driscoll & Kraay, 1998) standard error estimation to address these estimation issues. As presented in Table 7, the regression coefficients of *Transformation* are significantly positive, showing hypothesis 1 still holds.

To ensure the robustness of the mechanism analysis, we also

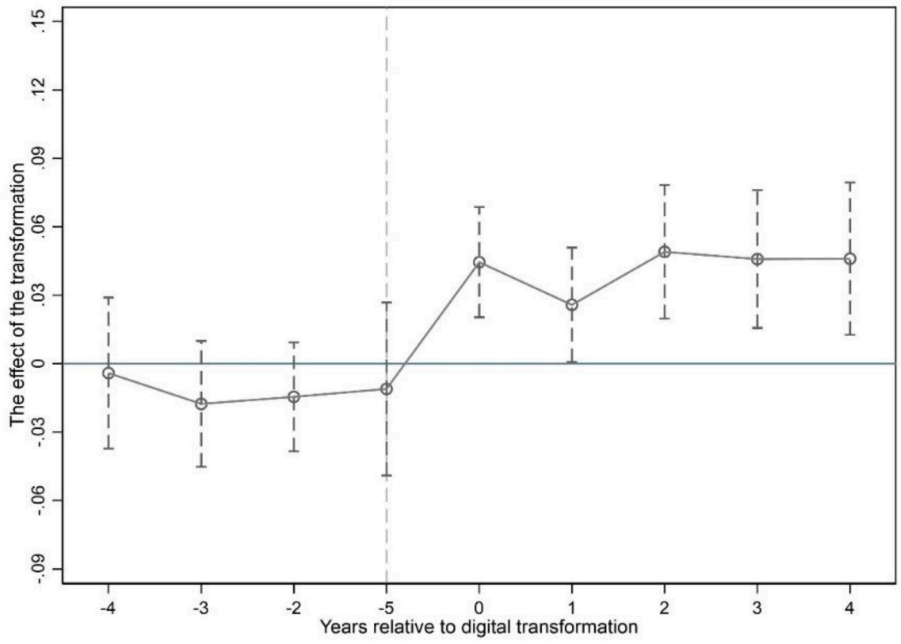


Fig. 3. Parallel trend test.

Table 5
Robustness test by using GMM method.

Variables	TFP			Emcsr	Pucsr
	(1)	(2)	(3)	(4)	(5)
L. TFP	0.1958	0.2003	0.1930		
	(0.011)	(0.009)	(0.013)		
L. Emcsr				0.5326	
				(0.004)	
L. Pucsr					0.3431
					(0.046)
Transformation	0.0304	0.2996	0.0300	0.0695	0.1279
	(0.000)	(0.000)	(0.000)	(0.017)	(0.032)
Emcsr		0.0039			
		(0.054)			
Pucsr			0.0001		
			(0.987)		
Other variables	Control	Control	Control	Control	Control
Arellano-Bond	−10.51	−10.53	−10.55	−3.93	−3.31
test for AR(1)	[0.000]	[0.000]	[0.000]	[0.000]	[0.001]
Arellano-Bond	−1.41	−1.39	−1.43	−1.36	1.31
test for AR(2)	[0.159]	[0.166]	[0.154]	[0.173]	[0.190]
Hansen J test	178.96	177.51	175.70	24.62	45.97
	[0.133]	[0.137]	[0.159]	[0.136]	[0.123]
Observations	14334	14334	14334	14334	14334

Note:The values in parentheses are P>|z|, and the values in brackets are Prob > chi2.

conducted a Sobel-Goodman mediation analysis. The tests for the *Emcsr* and *Pucsr* variables yield Z-values of 3.039 and 2.904 and meet the 5 % significance threshold, confirming our results' reliability. The findings from the Sobel method aligned with the above analyses, demonstrating the robustness of the mechanism tests.

4.4. Endogeneity test

Considering that selection bias may have a negative impact on parameter estimation in the model, Propensity Score Matching (PSM) is a causal inference method that can effectively control and eliminate selection bias through propensity value matching, thereby reducing the influence of endogeneity on the conclusions to be proven in this study. When conducting PSM analysis, the first step is to calculate the propensity score for each company. The propensity score is a measure of the likelihood that a company will receive a treatment (in this case, digital transformation). The specific process for calculating the propensity score is as follows: Firstly, identify the treatment and control groups in the study, namely companies that have implemented digital transformation and those that have not. Next, collect a range of characteristic variables for the companies, including *size*, *lev*, *ato*, *growth*, *dual*, *firmage*, *balance*, *inst*, *mfee*, *roe*, *audit*. Following this, develop a Probit regression model to forecast the likelihood of a company undergoing digital transformation. In this model, the aforementioned characteristic variables serve as independent variables, while the binary outcome of whether the company has undergone digital transformation acts as the dependent variable. Leveraging the values of these characteristic variables, the model is employed to compute a propensity score for each company. A higher propensity score signifies an increased probability of the company embracing digital transformation. Ultimately, match the treatment and control groups based on their propensity scores, striving for maximum similarity between the two groups in terms of their characteristic variables. This step is crucial for minimizing the influence of confounding factors, thereby facilitating a more precise evaluation of the impacts of digital transformation.

This study used nearest-neighbor matching and radius caliper matching for validation. Adopting both matching methods for analysis ensures more reliable test results. First, a 1:1 nearest neighbor matching was used to match study samples, followed by radius caliper matching. The caliper range was set at 0.01, which means that observations with propensity score differences of 1 % were matched. After propensity score matching, the majority of observed values fall within the common range (on support), resulting in a minimal loss of samples. After nearest-neighbor matching, the standardized bias of variables was less than 10 %, and the *t*-test did not reject the null hypothesis. Besides, the standardized bias of most variables decreased in Fig. 4. These indicate that the matching method has achieved a good data balance.

Table 8 below shows the test results for the average treatment effect. For nearest neighbor matching, the average treatment effect on the treated (ATT) is 4.92, with a corresponding t-value of 10.80, which is statistically significant at a 99 % confidence level. For Radius caliper matching, the ATT estimate is 5.73 and is statistically significant as a positive value at a 1 % significance level. Both PSM tests indicate that ATT is positive and still support the hypothesis.

To further ensure the reliability of research results, we conducted regression analysis using matched samples. The variable "*Transformation*" reflects whether the company has implemented transformation. The regression results of the impact of the transformation on TFP are shown in Table 9. Based on the matched samples using nearest neighbor matching, the regression results show a coefficient of 0.074 for *Transformation*, with a T-value of 8.19. Similarly, based on the samples using radius caliper matching, the results show a coefficient value of 0.073 for *Transformation*, with a T-value of 11.29, which is also positive

Table 7
Regression results based on the Driscoll-Kraay standard error estimation.

Variables	*(1)TFP_lp*	*(2)TFP*_ols	*(2)TFP*_gmm	*(4)TFP*_fe
Transformation	0.027***	0.019***	0.037***	0.018***
	(5.83)	(5.08)	(4.67)	(5.04)
Others	control	control	control	control
Constant	−2.249***	−3.451***	0.670**	−3.799***
	(-4.71)	(-4.90)	(2.74)	(-4.98)
Year	Fixed	Fixed	Fixed	Fixed
Firm	Fixed	Fixed	Fixed	Fixed
Observations	17032	17032	17032	17032
Within R-sq	0.5836	0.6765	0.2925	0.686
F-statistic	3491.72***	70669.81***	4513.49***	62545.55***

Note: (1) maximum lag 1; (2) ***p < 0.01, **p < 0.05, *p < 0.1.

Table 6
Regression results after changing the estimation method of TFP.

Variables	(1)*TFP-ols*	(2)*TFP-ols*	(3)*TFP*-gmm	(4)*TFP*-gmm	(5)*TFP*-fe	(6)*TFP*-fe
Transformation	0.019***	0.018***	0.037***	0.083***	0.018***	0.013***
	(5.47)	(6.65)	(9.14)	(20.91)	(5.00)	(4.76)
Others	control	control	control	control	control	control
Year	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
Industry	No	Fixed	No	Fixed	No	Fixed
Firm	Fixed	No	Fixed	No	Fixed	No
Constant	−3.451***	−7.961***	0.670***	−0.241**	−3.799***	−8.620***
	(-21.58)	(-117.46)	(3.59)	(-2.43)	(-23.23)	(-124.48)
Observations	17032	17032	17032	17032	17032	17032
R-squared	0.677	0.919	0.293	0.455	0.686	0.925
F-statistic	1379.25***	13302.18***	272.61***	934.60***	1443.24***	14304.33***

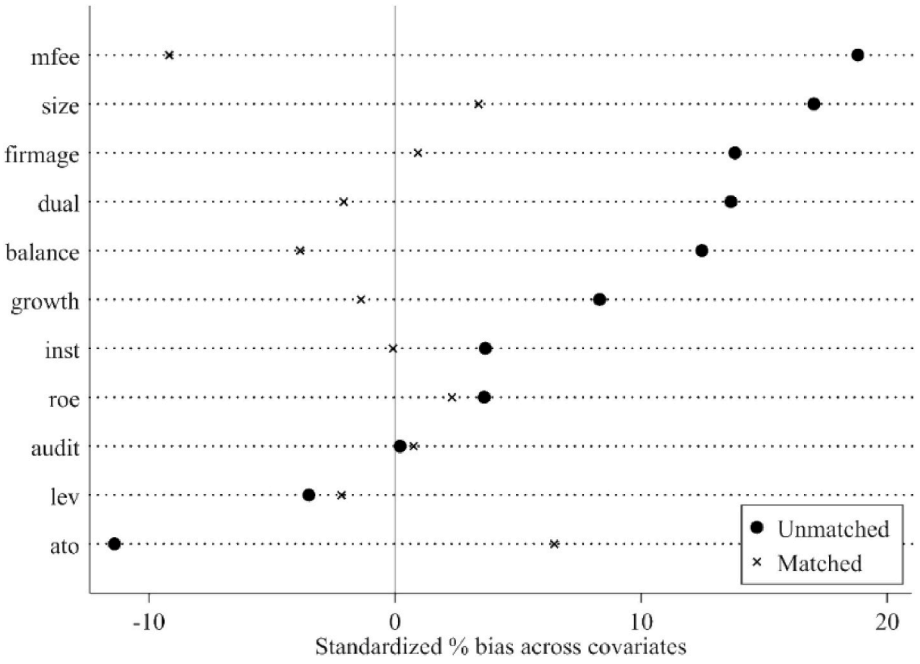


Fig. 4. Standardized % bias across covariates for nearest neighbor matching.

Table 8
Test results based on PSM method.

Variable	(1)Nearest neighbor matching			(2)Radius caliper matching		
	Difference	S.E.	T stat	Difference	S.E.	T stat
TFP Unmatched	0.1800	0.01667	10.80	0.1800	0.0167	10.80
ATT	0.1600	0.0325	4.92	0.1469	0.0256	5.73
ATU	0.0429	.	.	0.0515	.	.
ATE	0.0967	.	.	0.0953	.	.

Note: S.E. does not take into account that the propensity score is estimated.

Table 9
Regression results based on PSM.

Variables	*(1) Nearest neighbor matching*		*(2) Radius caliper matching*	
	Coefficient	T-value	Coefficient	T-value
Transformation	0.074***	8.19	0.073***	11.29
size	0.648***	113.12	0.651***	191.80
lev	−0.024	−0.81	0.022	1.24
ato	1.091***	72.09	1.065***	120.35
growth	0.062***	5.36	0.058***	8.20
dual	0.005	0.50	−0.002	−0.26
firmage	−0.016	−1.08	−0.015*	−1.76
balance	−0.016**	−2.08	−0.013***	−2.70
inst	0.122***	5.58	0.081***	6.01
mfee	−1.237***	−14.04	−1.135***	−20.96
roe	0.024	0.67	0.149***	6.90
audit	0.059**	2.30	0.049***	3.17
Constant	−6.143***	−46.66	−6.154***	−79.14
Year	Fixed	–	Fixed	–
Industry	Fixed	–	Fixed	–
Observations	5988	–	15,101	–
R-squared	0.8848	–	0.8915	–

at the 1 % significance level. In conclusion, these empirical results using the PSM method still support the hypothesis proposed in this paper.

In conclusion, the test results obtained by different methods all support our proposed hypotheses.

5. Discussion

5.1. Interpretation of findings

This study provides significant insights into the relationship between digital transformation, corporate productivity, and the role of employee and public responsibility within Chinese manufacturing companies. Our results support hypothesis 1, which suggests that digital transformation directly enhances TFP. This aligns with previous research findings that emphasize the efficiency gains and performance improvements brought about by the adoption of digital technologies (Chen et al., 2016; Lu et al., 2024; Luo et al., 2018; Sousa-Zomer et al., 2020; Zhai et al., 2022). This may indicate that digital technology can strengthen the allocation of production resources and lead to better utilization of corporate resources (Ding, Vuković, et al., 2024; Ferreira et al., 2019; Heavin & Power, 2018; Jiang & Li, 2024).

We analyze in depth the mechanisms between digital transformation and TFP. On the one hand, our study reveals that digital transformation indirectly boosts TFP by promoting the fulfillment of employee responsibility. As some researchers have argued, digital transformation enhances CSR by integrating resources, improving human capital efficiency, and expanding employment opportunities (Jiang & Li, 2024; Sun et al., 2024). Furthermore, as a type of production resource, the development and utilization of human resources may facilitate the enhancement of TFP (Ding, Vuković, et al., 2024). Our finding support the view that employee responsibility is pivotal in leveraging digital transformation to boost TFP. On the other hand, digital transformation indirectly enhances TFP by strengthening the fulfillment of public

responsibility. Public responsibility plays an indispensable role in consolidating stakeholder relationships, acquiring key resources, and enhancing corporate overall performance (Wang and Qian, 2011). Actively fulfilling public responsibilities helps establish a positive image and reputation for the company, making various stakeholders more willing to cooperate and provide production resources with companies. Digital transformation can greatly improve the efficiency of public responsibility fulfillment, making it easier for companies to obtain production factors. Besides, as the transformation deepens, information-sharing mechanisms are established, thereby reshaping the pattern of resource allocation. These changes not only enrich the supply of production factors but also further optimize resource allocation, ultimately driving the improvement of TFP. In summary, this dual pathway is crucial, as it indicates that responsible digitalization, where companies prioritize social responsibilities alongside technological advancements, can lead to superior productivity outcomes. This aligns with the notion that socially responsible practices contribute to better economic performance (Tang and Yang, 2022).

Digital technology can serve as a double-edged sword, either enabling corporations to better fulfill their social responsibilities or fostering irresponsible digitalization practices (Alexopoulos & Cohen, 2016; Bisht et al., 2023; Cardinali & De Giovanni, 2022; DeCanio, 2016; Kong & Liu, 2023). Our research reveals that digital technology enhances TFP by improving corporate commitment to employee responsibilities. This finding aligns with the perspective advanced by Nadeem and Selimović, among others, that the application of digital technology holds the potential to bolster job engagement (Nadeem et al., 2024; Selimović et al., 2021). In the context of Chinese manufacturing companies, digital transformation predominantly exerts a positive influence on employment, contrasting with the scenario observed in Germany, where it has been associated with the reduction of low-skilled jobs, decreased employment opportunities, and heightened unemployment risks (Dengler & Matthes, 2018).

Our findings support previous research indicating that digital technologies can enhance productivity (Cheng et al., 2023; Ding, Sheng, et al., 2024; Jin et al., 2023; Relich, 2017; Wang et al., 2023a). This confirms that digital transformation has the potential to drive efficiency, thereby ultimately improving TFP. By modernizing processes and integrating advanced technologies, manufacturing companies can achieve significant gains in output and operational efficiency. These outcomes are consistent with the endogenous growth theory, emphasizing technological progress as a crucial driver of sustained economic growth (Romer, 1986, 1990). In addition, our results reveal that fulfilling employee and public responsibilities serves as a crucial pathway for enhancing TFP. This perspective extends existing literature by highlighting the importance of a balanced approach that combines digital advancement with social responsibility. Previous studies often focused on environmental and economic outcomes of digital transformation (Alkaraan et al., 2022; Chen & Hao, 2022; Ding, Sheng, et al., 2024; Ghobakhloo et al., 2023; Lin & Zhang, 2023; Zhao et al., 2024), frequently overlooking social dimensions. Our findings indicate that responsible digitalization, which includes commitments to employees and the public, can amplify the positive impacts of digital transformation on productivity, offering a fresh perspective for future research and practical strategies.

The methodology used in this study is scientific and has advantages in addressing the mechanisms. When examining the mechanism, we mainly used fixed effects model regression, as well as PSM, DID, and GMM methods. The fixed effects model has advantages such as addressing omitted variable issues, providing more individual dynamic information, and controlling for individual heterogeneity. PSM can solve selection bias problems and reduce the impact of omitted variables on research results. DID not only mitigates omitted variable bias, but also largely avoids the problem of endogeneity (Ding, Sheng, et al., 2024). In addition, GMM estimation has properties such as robustness and consistency (Du et al., 2023). Under appropriate conditions, the results of GMM estimation can gradually converge to the true parameter values as the sample size increases. Across different methods, parameter estimates all support the research hypothesis. Besides, we mainly referred to research by Alberto Alesina (Alesina & Zhuravskaya, 2011) for mechanism analysis and also performed a Sobel-Goodman mediation analysis. Both methods confirm the robustness of the mechanism tests. The methods used in this paper have their own advantages, and the combination of these methods provides strong support for robust research results.

5.2. Theoretical implications

The focus of endogenous growth theory has been on the influence of technological advancements on economic gain. Yet, it has not delved deeply into the link between enterprise productivity and digital transformation. Related researchers have also discussed the link between the digital economy and single-factor productivity (Hao et al., 2023; Lyu et al., 2023; Pan et al., 2022), but there is a lack of explanation about the effectiveness of digital transformation on TFP gains. This paper shows that digital transformation enhances company TFP, answering the substantive question about digital transformation's effectiveness and expands the explanatory scope of endogenous growth theory, thus making an innovative contribution to the existing literature.

Industry 4.0, a subset of digital transformation (Erol et al., 2016), may threaten employee responsibility (Horváth & Szabó, 2019). However, Cardinali and Giovanni (Cardinali & De Giovanni, 2022) argued that the combination of digital technologies enables companies to achieve responsible digitalization. There is also a traditional view that fulfilling social responsibility consumes and encroaches upon company's production resources, potentially harming the company (Aupperle et al., 1985). Therefore, how digital transformation affects CSR is controversial, and it is unclear in the literature whether responsible digitalization contributes to improving corporate TFP. Our empirical study finds that digital transformation contributes to improving corporate employee and public responsibility, and it also enhances TFP by promoting the fulfillment of employee and public responsibility. These empirical results support the views of Cardinali and Giovanni (Cardinali & De Giovanni, 2022) and provide us with two points of understanding: first, digital transformation enables firms to fulfill CSR, and it is more of an opportunity than a threat to fulfilling CSR; second, fulfilling employee responsibilities and public responsibilities during digital transformation will contribute to improving TFP. This paper addresses academic controversy regarding CSR's impact on TFP, contributing to the theory of responsible digitalization, and offering new insights into the complex interplay between technology and social responsibility. At the company level, this study finds that employee and public responsibilities serve as pathways for improving TFP during digital transformation, enhancing the understanding of the mechanism about digital transformation affects corporate production efficiency.

5.3. Practical implications

Based on the research findings, businesses that do not promptly adopt digital strategies will find it challenging to stay competitive in today's digital age. To enhance TFP, companies should prioritize leveraging digital transformation to improve corporate efficiency and effectiveness. This involves integrating digital technologies such as cloud computing, big data analytics, and artificial intelligence into core business processes. By doing so, companies can streamline operations, reduce costs, and improve decision-making capabilities. To implement this strategy, companies should allocate sufficient resources to digital initiatives, including funding, personnel, and training. Additionally, they should establish a clear digital transformation roadmap with specific goals, timelines, and milestones to ensure successful implementation.

To effectively leverage digital transformation, companies can prioritize building a pool of digital talent. This involves attracting and retaining individuals with specialized skills in areas such as data science, cybersecurity, and digital marketing. Companies should offer competitive salaries, benefits, and career development opportunities to attract top talent. Additionally, they should safeguard employees' rights and interests during digital transformation by providing clear communication, support, and training. Fulfilling the company's social responsibility towards employees will not only enhance employee satisfaction and loyalty but also ensure the effective role of talent and human resources in enhancing TFP. To further strengthen employee responsibility, companies should encourage a culture of continuous learning and innovation, fostering an environment where employees feel empowered to contribute to the company's digital transformation efforts.

In addition to focusing on digital transformation and employee responsibilities, companies should also pay attention to their public responsibilities. Fulfilling public responsibilities can promote corporate TFP. To fulfill these responsibilities, companies should actively pay taxes and comply with all relevant laws and regulations. Furthermore, they should engage in public welfare and donations to support communities and social causes. By actively participating in social welfare activities, companies can demonstrate their commitment to corporate social responsibility and strengthen their relationships with stakeholders, including customers, employees, and the wider community.

6. Conclusion

This paper selects Chinese manufacturing companies as research samples and answers the two research questions by developing hypotheses and empirical tests, obtaining the three findings.

First, this study demonstrates that digital transformation improves the TFP of manufacturing companies. Second, the study reveals that digital transformation enhances TFP of manufacturing companies by promoting the fulfillment of employee and public responsibilities. Both employee responsibility and public responsibility play a promoting role. Being responsible to employees and the public during digital transformation helps to boost corporate TFP. Third, the study indicates that while digital transformation indirectly boosts TFP by reinforcing employee and public responsibilities, its primary effect on TFP is direct.

This study proposes a novel theoretical framework of "responsibility digitalization," explaining how digital transformation enhances corporate productivity by promoting companies' fulfillment of responsibilities towards employees and the public. To our knowledge, this is the first study to find that integrating employee and public responsibilities into digital strategies boosts corporate productivity. The originality of this study lies at the intersection of digital transformation and corporate social responsibility, particularly in the context of employee and public responsibility. This fresh perspective could significantly influence future research and practical strategies in the field of responsible digitalization. By filling a critical gap in existing literature, this study offers actionable insights for policymakers and industry leaders aiming to balance technological advancements with social responsibilities. The findings have the potential to drive future research and inform strategic decisions in the broader context of digital transformation.

6.1. Limitations and future research trends

The present study has its limitations. This paper uses a novel approach to quantify the digital transformation. Although this method reflects scientific and rationality, this quantitative method is more complicated to operate. The authors have not proposed a more scientific method to quantify the intensity of digital transformation.

Future research can explore the following areas more deeply. First, this study only discusses the role of certain aspects of Corporate Social Responsibility (CSR) in the digital transformation process affecting TFP. The mechanism might be more complex, requiring further exploration of additional mechanism variables. Second, future research can expand the sample to include more regions, particularly those at different stages of development, to verify the universality and variability of digital strategy's impact on corporate productivity. Third, while this study focuses on the manufacturing industry, the digital transformation process and its impact may differ across various industries. Future research can explore the role and effects of digital transformation in different sectors to understand both the differences and commonalities. Fourth, future research can investigate the specific impacts of different digital transformation pathways and strategies, such as smart manufacturing, data-driven decision-making, and supply chain digitalization, thus offering more actionable guidance for companies in developing their digital strategies. Fifth, research indicates that digital transformation indirectly enhances firm productivity by fulfilling responsibilities towards employees and the public. Future studies can delve deeper into the specific mechanisms involved. For instance, examining how factors like employee engagement, skill enhancement, and job satisfaction play roles in the transformation process, and how companies can improve public image and market competitiveness through social responsibility activities. Sixth, digital transformation is a dynamic process, and its impact on firms' TFP can change over time. Future research could adopt longitudinal study designs to analyze the long-term effects of digital transformation at different stages on firm productivity, helping companies better understand and manage the dynamic process of digital transformation. Additionally, digital transformation is influenced not only by internal factors but also by external environments and policies. Future research could investigate how external factors such as policy support, market circumstances, and technological infrastructure in different countries and regions affect the outcomes of digital transformation, thereby providing references for policymakers. By exploring these future research directions in depth, a more comprehensive understanding of the multifaceted impact of digital transformation on corporate productivity can be achieved. This will provide scientific evidence and practical guidance for the sustained development of enterprises in the digital era.

Credit author statement

Xiangan Ding: Writing original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. Andrea Appolloni: Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. Mohsin Shahzad: Conceptualization. Yue Liu: Review & editing. Shaojie Han: Review & editing.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Ethical statements

Not applicable

Funding

This research was funded by:

(1) National Natural Science Foundation of China (72204104)
(2) Liaoning Normal University High-Level Scientific Research Achievement Cultivation Project (24GDW002)
(3) National Natural Science Foundation of China (72102028)

Declaration of competing interests

The authors declare no competing interests.

Acknowledgments

Prof. Andrea Appolloni acknowledges the funding of this study supported by the European Union - NextGenerationEU, in two projects: 1) the first project in the framework of the GRINS - Growing Resilient, Inclusive and Sustainable project (GRINS PE00000018 – CUP E83C22004690001); 2) the second project in the framework of the ECS 0000024 Rome Technopole – CUP B83C22002820006, NRP Mission 4 Component 2 Investment 1.5.

Data availability

The datasets analyzed during the current study are available from the corresponding author on reasonable request.

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